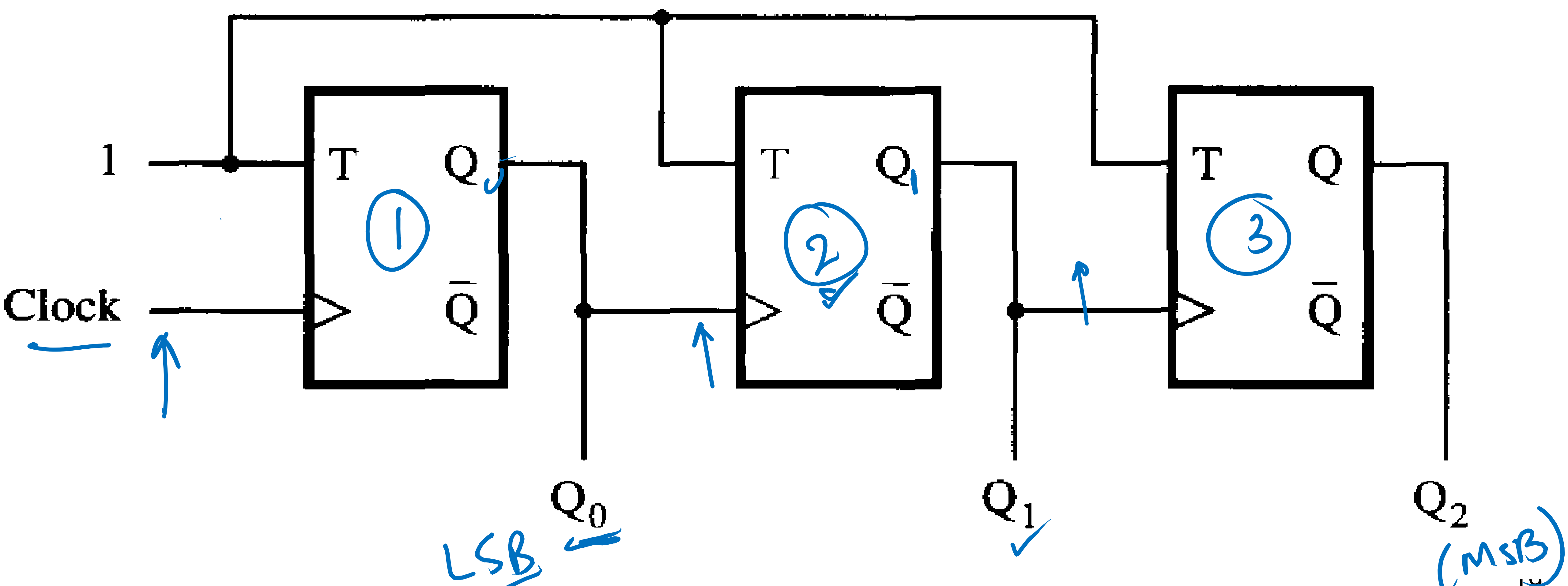
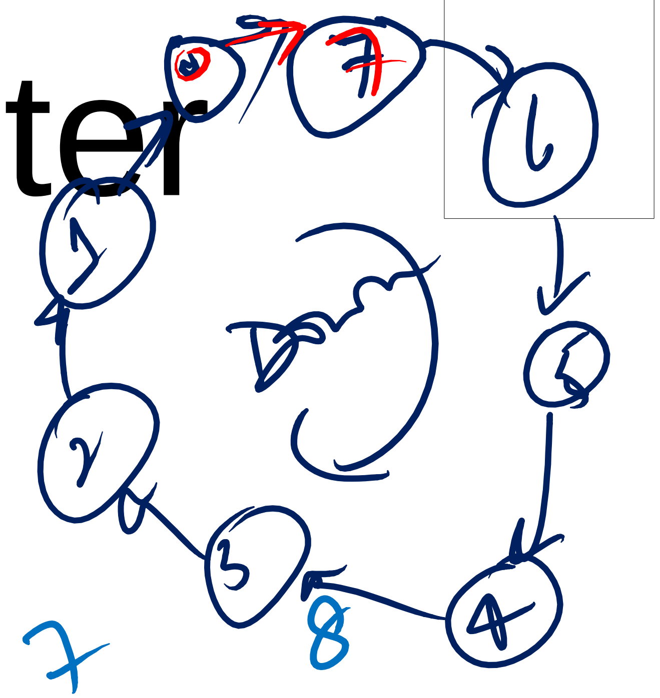
Asynchronous Counter - Down Counter

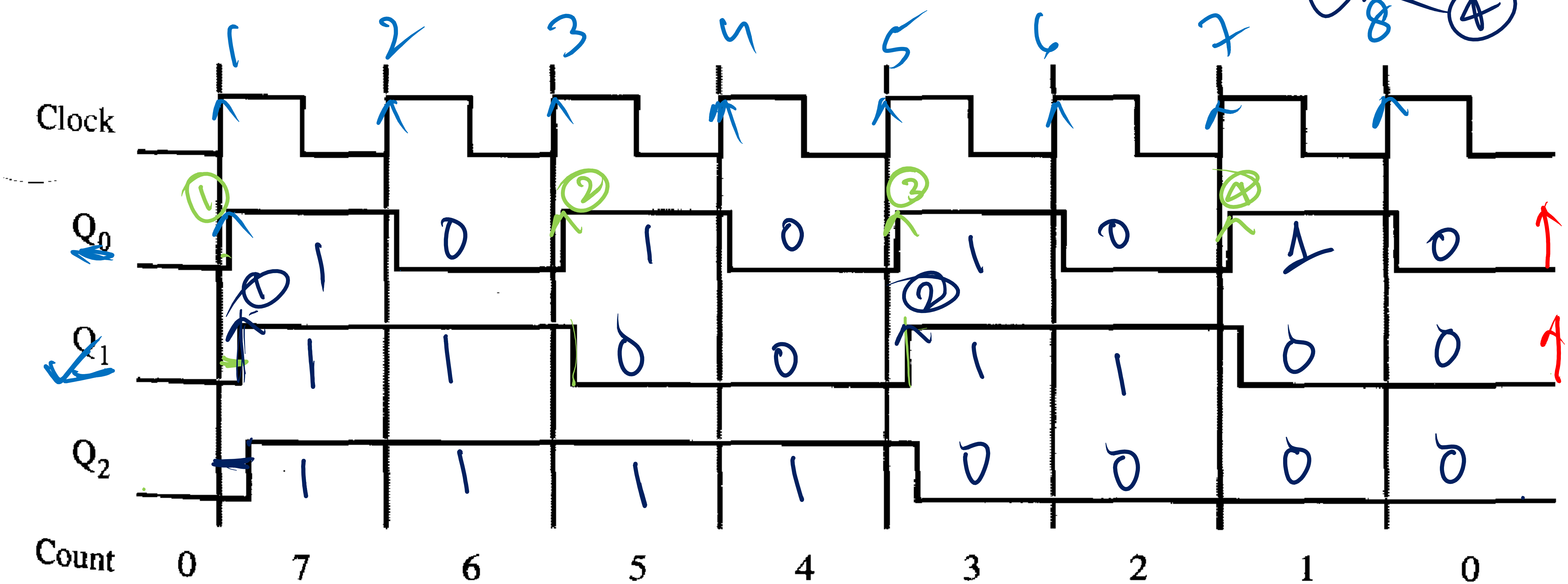
Divide by 8
mod 8
3 bit

- The only difference is that in the clock inputs of the second and third flip- flops are driven by the Q outputs of the preceding stages, rather than by the Q' outputs.



Asynchronous Counter - Down Counter





Synchronous Counter

100ns

5ns

clock pulse

1 FF 50ns delay

- The asynchronous counters are simple, but not very fast
- If a counter with a larger number of bits is constructed in this manner, then the delays caused by the cascaded clocking scheme may become too long to meet the desired performance requirements.
- We can build a faster counter by clocking all flip-flops at the same time. This kind of counter is called as synchronous counter

Synchronous Counter

- Q0 changes on each clock cycle
- Q1 changes only when Q0 = 1
- Q2 changes only when both Q1 and Q0 are equal to 1

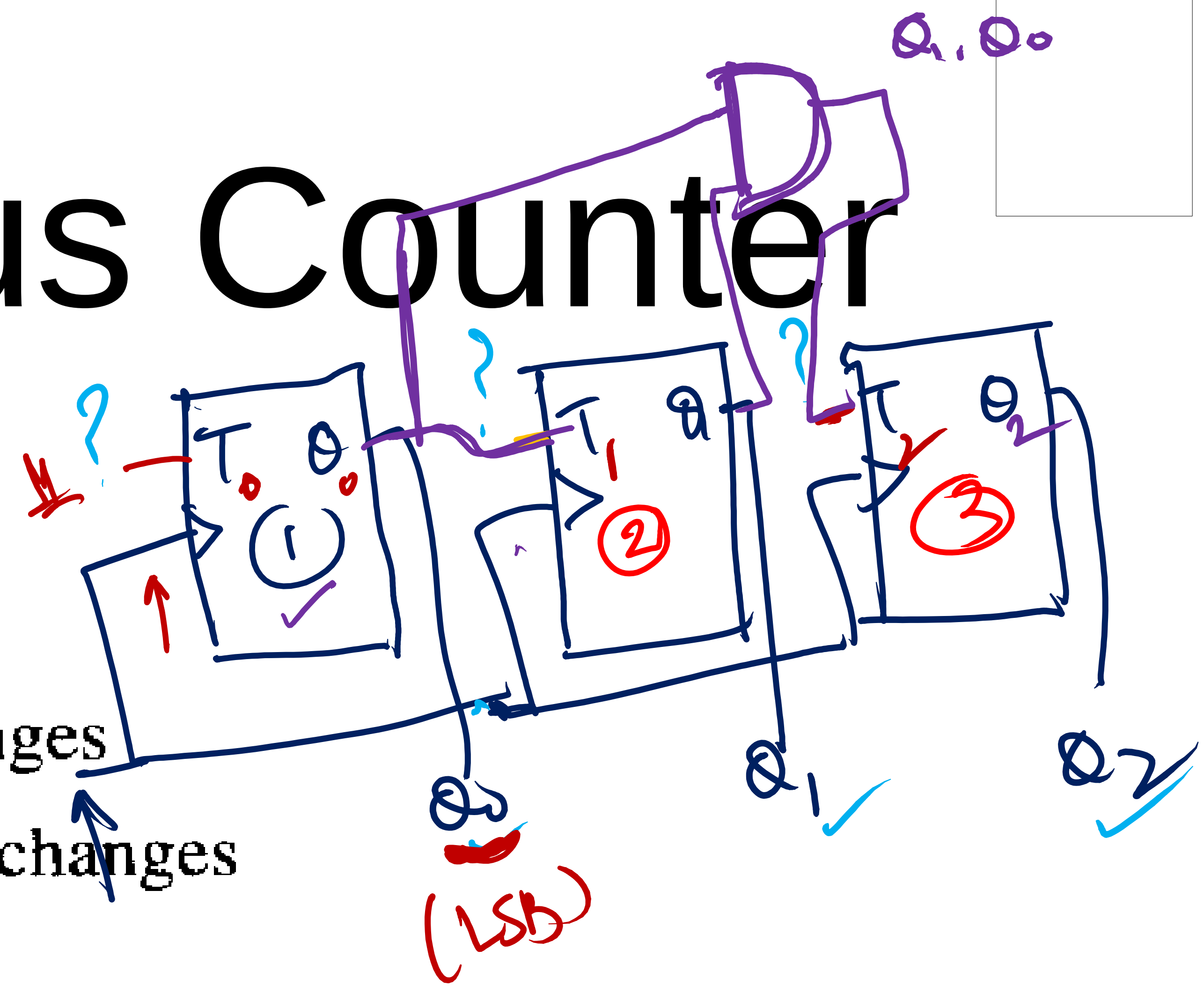
Clock cycle	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Q_1 changes

Q_2 changes

$$T_0 = 1$$
$$T_1 = Q_0$$
$$T_2 = Q_0 Q_1$$
$$T_3 = Q_0 Q_1 Q_2$$
$$\vdots$$
$$T_n = Q_0 Q_1 \cdots Q_{n-1}$$

Synchronous Counter



Clock cycle	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Q_1 changes

Q_2 changes

2 1 LSB

Q_0 toggles everytime $T_0 = 1$

Q_1 toggles when $Q_0 = 1$ $\Rightarrow T_1 = Q_0$

Q_2 toggles when $Q_0 = Q_1 = 1$ $T_2 = Q_1 \cdot Q_0$

$T_3 = Q_2 . Q_1 . Q_0$

Q_3 :

Q_n toggles $Q_{n-1} = Q_{n-2}$ ...

$T_4 = Q_3 . Q_2 . Q_1 . Q_0 = Q_3 \cdot T_3$

$T_n = Q_{n-1} \cdot Q_{n-2} \cdots Q_0$

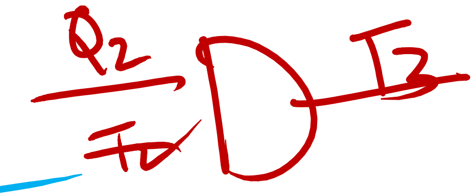
Synch. up counter

$T_0 = 1$

$T_1 = Q_0$

$T_2 = Q_1 \cdot Q_0 = Q_1 \cdot T_1$

$T_3 = Q_2 \cdot Q_1 \cdot Q_0 = Q_2 \cdot T_2$



$T_4 = Q_3 \cdot Q_2 \cdot Q_1 \cdot Q_0 = Q_3 \cdot T_3$

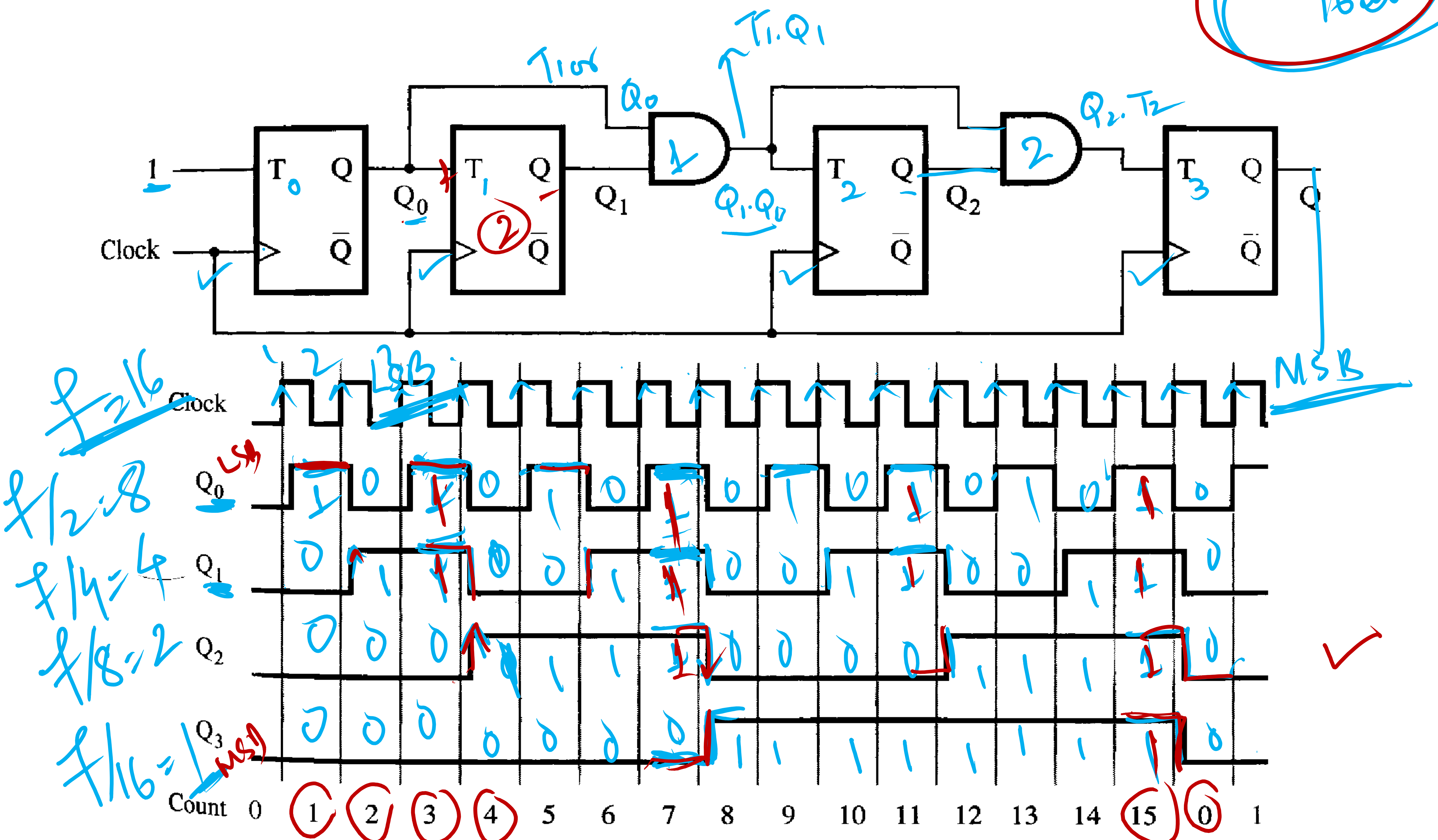
$\vdots$

$T_n = Q_{n-1} \cdot Q_{n-2} \cdots Q_0 = Q_{n-1} \cdot T_{n-1}$

Synchronous Counter

4 bit up counter MOD 16

Divide by 16 counter



Synchronous Counter with D Flip-Flops

- While the toggle feature makes T flip-flops a natural choice for the implementation of counters, it is also possible to build counters using other types of flip-flops.

- The JK flip-flops can be used in exactly the same way as the T flip-flops because if the J and K inputs are tied together, a JK flip-flop becomes a T flip-flop.

- D flip-flops can also be used, by converting T flip-flop to D flip-flop

Synchronous Counter with D Flip-Flops

$$D = Q\overline{T} + \overline{Q}T$$

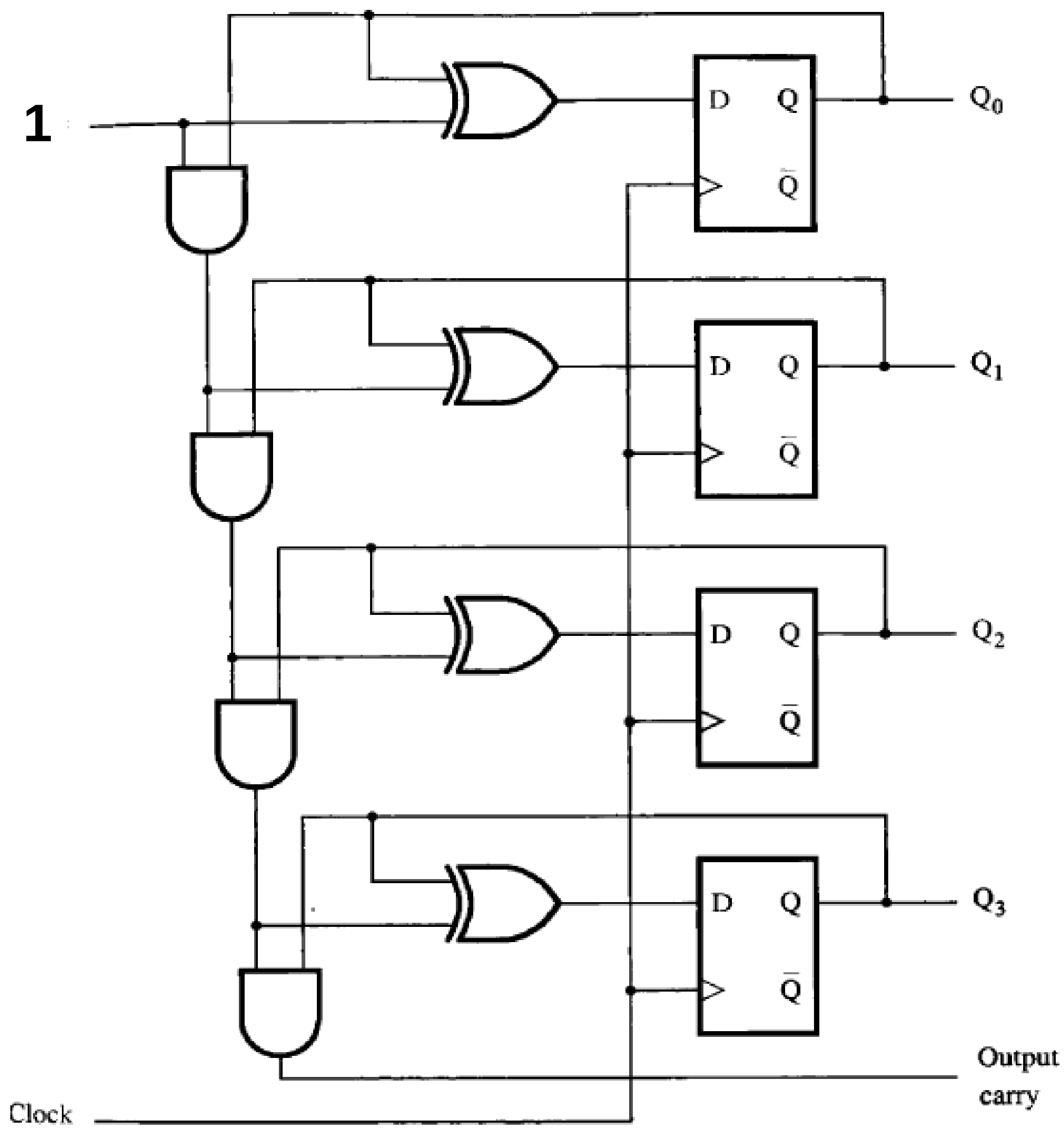
$$= Q \oplus T$$

$$D_0 = \overline{Q}_0 = 1 \oplus Q_0$$

$$D_1 = Q_1 \oplus Q_0$$

$$D_2 = Q_2 \oplus Q_1 Q_0$$

$$D_3 = Q_3 \oplus Q_2 Q_1 Q_0$$



Ring Counter

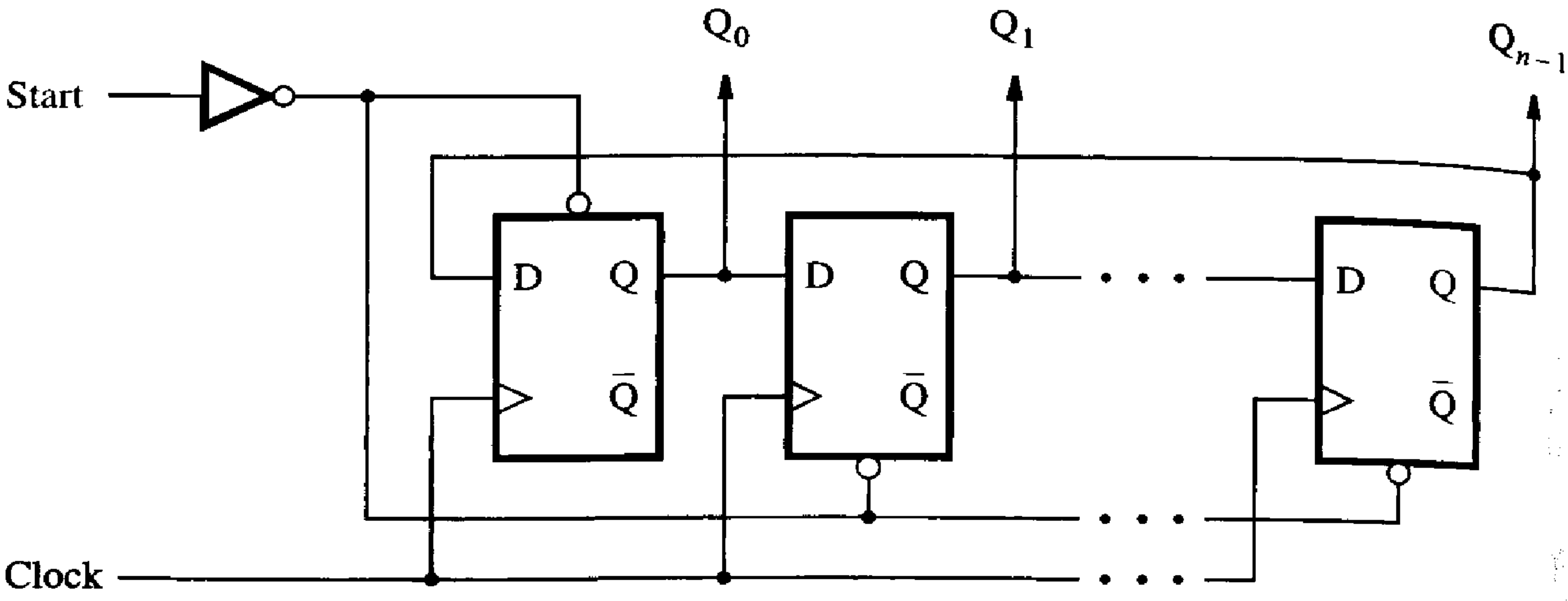
- It is possible to devise a counter like circuit in which each flip-flop reaches the state Qi == 1 for exactly one count, while for all other counts Qi = 0.

- Then Qi indicates directly an occurrence of the corresponding count.

- Since this does not represent binary numbers, it is better to say that the outputs of the flips-flops represent a code.

- Such a circuit can be constructed from a simple shift register,

Ring Counter

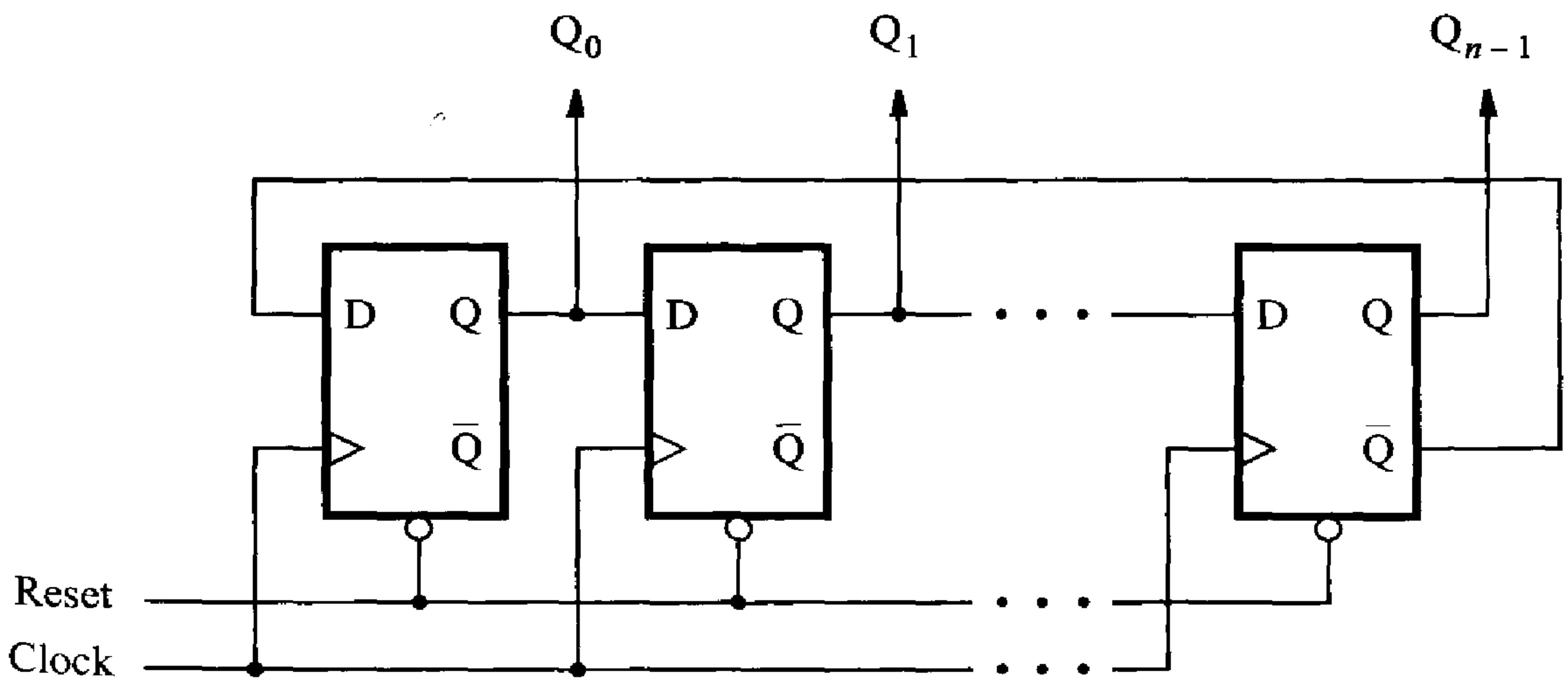


Ring Counter

- The Q output of the last stage in the shift register is fed back as the input to the first stage, which creates a ring like structure.
- If a single 1 is injected into the ring, this 1 will be shifted through the ring at successive clock cycles.
- For example, in a four-bit structure, the possible codes Q0Q1Q2Q3 will be 1000,0100,0010, and 0001.
- Such encoding, where there is a single 1 and the rest of the code variables are 0, is called a one-hot code

Johnson Counter

- An interesting variation of the ring counter is obtained if, instead of the Q output, we take the Q output of the last stage and feed it back to the first stage



Johnson Counter

- An n-bit counter of this type generates a counting sequence of length 2n.
- For example, a four-bit counter produces the sequence 0000, 1000,1100,1110,1111, 0111, 0011, 0001,0000, and so on.
- Note that in this sequence, only a single bit has a different value for two consecutive codes.

Thank You !!!